

Método de superposición

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Análisis de circuito con superposición

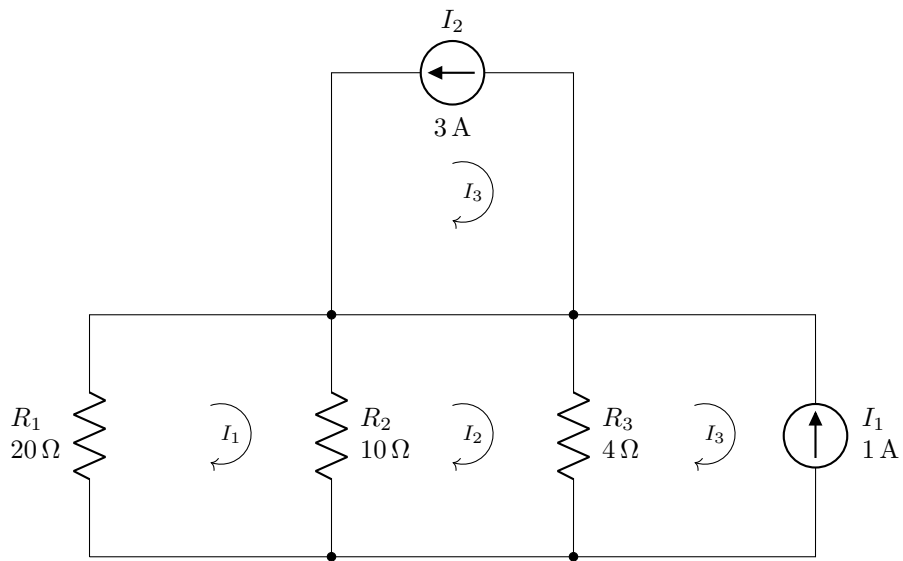


Figure 1:

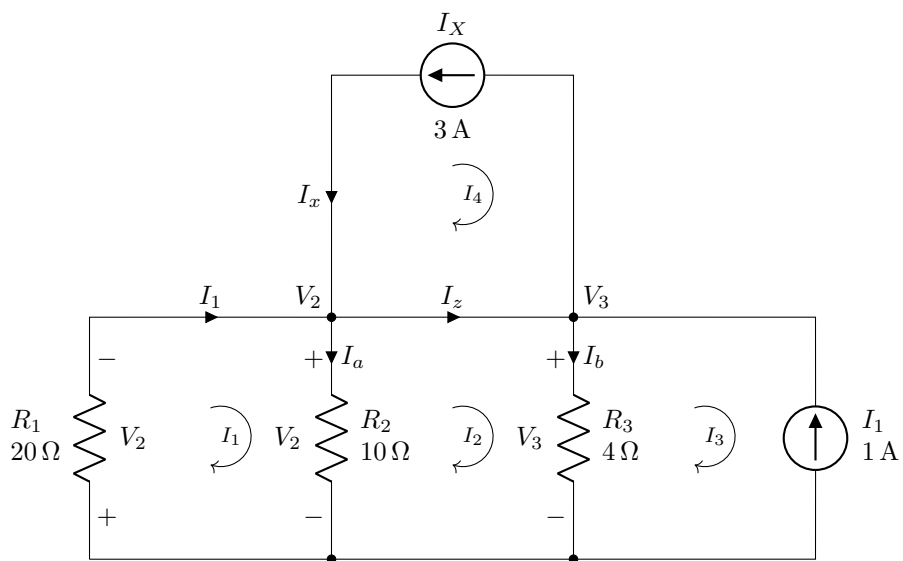


Figure 2:

$$V_{out} = V_{FB} \times \left(1 + \frac{R_1}{R_2}\right) = 0.6 \text{ V} \times \left(1 + \frac{R_1}{R_2}\right)$$

If $R_1 = 330 \text{ k}\Omega$ and $R_2 = 68 \text{ k}\Omega$ then:

$$V_{out} = 0.6 \text{ V} \times \left(1 + \frac{330 \text{ k}\Omega}{68 \text{ k}\Omega}\right) = 3.51 \text{ V}$$

$$V_{out} = 0.6 \text{ V} \times \left(1 + \frac{249 \text{ k}\Omega}{68 \text{ k}\Omega}\right) = 2.797 \text{ V}$$

$$R_2 = \frac{V_{FB}}{I_{FB}} = \frac{0.6 \text{ V}}{5 \mu\text{A}}$$

$$I_{FB} = \frac{V_{FB}}{R_2} = \frac{0.6 \text{ V}}{68 \text{ k}\Omega} = 8.82 \mu\text{A}$$

$$R_1 = R_2 \times \left(\frac{V_{out}}{V_{FB}} - 1\right) = R_2 \times \left(\frac{V_{out}}{0.6 \text{ V}} - 1\right)$$

$$R_1 = 68 \text{ k}\Omega \times \left(\frac{3.3 \text{ V}}{0.6 \text{ V}} - 1\right) = 306 \text{ k}\Omega$$

$$R_2 = \frac{V_{FB}}{I_{FB}} = \frac{0.6 \text{ V}}{5 \mu\text{A}} = 120 \text{ k}\Omega \quad (1)$$

$$R_1 = R_2 \times \left(\frac{V_{out}}{0.6 \text{ V}} - 1\right) = 120 \text{ k}\Omega \times \left(\frac{3.3 \text{ V}}{0.6 \text{ V}} - 1\right) = 540 \text{ k}\Omega \quad (2)$$